API Reference

FastAPI REST service · Interactive Swagger UI at http://127.0.0.1:8000/docs · OpenAPI JSON at /openapi.json

Authentication

Method	Path	Purpose
POST	/api/auth/register	Create account; first account becomes administrator
POST	/api/auth/login	Password login; returns bearer token
POST	/api/auth/google	Google credential verification when configured
POST / GET / PATCH	/api/auth/logout · /me	Logout event; profile read/update

Students and analytics

Method	Path	Purpose
GET	/api/students	Search, filter, sort and paginate profiles
POST / PATCH / DELETE	/api/students[/{code}]	Create / update / delete (administrator for mutations)
GET	/api/students/{code}	Student details, timeline, recommendations
GET	/api/students/{code}/similar	Hybrid historical retrieval with metadata constraints
GET	/api/analytics/dashboard	Cohort stats, course trends, risk and intervention pipeline
GET	/api/courses	Course-level analytics
GET	/api/models/predict/{code}	Random forest demo predictions and holdout metrics

Assistant and support

Method	Path	Purpose
POST	/api/assistant/chat	Cohort question or student-ID analysis
POST	/api/agents/analyze	Performance/risk/support agents, similar cases, optional NIM summary/judge
POST	/api/recommendations	Advisor support suggestions for a student
GET / POST	/api/interventions	List/create plans
PATCH / DELETE	/api/interventions/{id}	Update/delete plan (owner/admin rules)
POST	/api/feedback	Save advisor rating/comment
GET	/api/analytics/evaluation	Feedback/evaluation configuration summary

Administration and utilities

Method	Path	Purpose
POST	/api/admin/ingest	Admin CSV metadata import; upload size cap applies
GET	/api/admin/audit	Administrator audit log
GET	/api/export/students.csv	Download student CSV
GET	/api/students/export.csv	Alternate CSV export
GET	/api/health	Unauthenticated service health

Auth, validation, and full schema

Attach Authorization: Bearer to protected routes. Input validation errors return 422; missing/invalid credentials return 401; role denial returns 403. Complete request/response models and parameters are exported to openapi.json. The backend also serves Swagger UI at /docs and ReDoc at /redoc.

Sample request / response: authentication

Request: POST /api/auth/login with JSON {email, password}. Response: {access_token, token_type: bearer, user: {name, email, role}}. Use the returned token as Authorization: Bearer <access_token> for protected calls.

Sample request / response: student filtering

Request: GET /api/students?course=AAA%20%C2%B7%202013J&level=HIGH&limit=2&offset=0 with a bearer token. Response: {items: [student profile objects], total: count, limit: 2, offset: 0}. Each profile includes score, clicks, risk score/band, completion, trends, and priority score.

Sample request / response: agent analysis

Request: POST /api/agents/analyze with JSON {student_code: OU00001_AAA_2013J}. Response includes summary, performance_analysis, risk_analysis, support_recommendation, agent_workflow, retrieved_cases, evidence, key_factors, and optional judge status.

Sample request / response: assistant

Request: POST /api/assistant/chat with JSON {message: Summarize the performance and engagement trends for course AAA 2013J}. Response includes summary, agent_workflow, evidence, similar_cases, recommended_intervention, limitations, and optional judge status.

Sample request / response: intervention CRUD

Create with POST /api/interventions and JSON {student_code, action, status}. Response includes {id, student_code, status, action}. Update with PATCH /api/interventions/{id}; delete with DELETE /api/interventions/{id}. Ownership or administrator permissions apply.

Role-scoped access management

The first account in an empty database is administrator; later registrations are advisors and cannot set their own role. GET /api/admin/users lists accounts and grants; PATCH /api/admin/users/{user_id} changes a role; POST /api/admin/access grants an advisor student ID or faculty course; DELETE /api/admin/access/{grant_id} revokes it. These changes are stored in the database. Advisor and faculty data is limited to grants; administrators see workspace records. GET /api/analytics/reports.csv returns course aggregates within the signed-in account scope (analysts receive cohort aggregates). Analysts cannot access student-level reports or identifiable endpoints.

All other request and response schemas

The endpoint inventory above covers all application routes, including Google authentication, profile CRUD, analytics, CSV, predictions, feedback, administration, and health. Exact required fields, parameter bounds, response media types, and validation schemas are available in deliverables/openapi.json and the running Swagger UI at /docs.